

JEE MAIN REVISION MASTER: ELECTROSTATICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 270 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Coulomb's Law, Electric Field & Dipole	217 PYQs	Forces & Fields
Very High	Gauss's Law & Electrostatic Potential	96 PYQs	Flux & Potential

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Coulomb's Law, Electric Field & Dipole	$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$, $E_{\text{axial}} = \frac{2kp}{r^3}$, $E_{\text{equatorial}} = \frac{kp}{r^3}$, $\vec{\tau} = \vec{p} \times \vec{E}$, $U = -\vec{p} \cdot \vec{E}$
Gauss's Law & Electrostatic Potential	$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$, $E = \frac{\sigma}{2\epsilon_0}$ (sheet), $E = \frac{\lambda}{2\pi\epsilon_0 r}$ (wire), $\vec{E} = -\nabla V$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Coulomb's Law, Electric Field & Dipole (Very High Yield — 217 PYQs)

Core Principle: Electrostatic forces, electric field intensity due to point charges and dipoles, and dipole torque/energy in uniform fields.

■ **Caution / JEE Pitfall:** $E_{\text{axial}} = 2E_{\text{equatorial}}$ for a short dipole. $\vec{E}_{\text{equatorial}}$ points opposite to dipole moment $\vec{p}$.

■ **Speed Trick:** Work done in rotating dipole from θ_1 to θ_2 : $W = pE(\cos\theta_1 - \cos\theta_2)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Two point charges $q_1 = 3\mu\text{C}$ and $q_2 = -4\mu\text{C}$ are placed at points $(2\hat{i} + 3\hat{j} + 3\hat{k})$ and $(\hat{i} + \hat{j} + \hat{k})$ res..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : In electrostatics, a co..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "Two short electric dipoles A and B having dipole moment p_1 and p_2 respectively are placed with their axis mutually perpendicular a..." [Ans: B]

3.2 Gauss's Law & Electrostatic Potential (Very High Yield — 96 PYQs)

Core Principle: Gauss's flux theorem, symmetry applications (infinite sheet, cylinder, sphere), and electric potential gradient $\text{vec}(\mathbf{E}) = -\text{vec}(\nabla)V$.

■ **Caution / JEE Pitfall:** Inside a charged hollow conducting sphere, electric field is ZERO, but potential is NON-ZERO and constant: $V_{\text{(inside)}} = (kq / R)$.

■ **Speed Trick:** Electric field lines are always perpendicular to equipotential surfaces and point in the direction of steepest potential decrease.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : In electrostatics, a co..." **[Ans: B]**
- **[2026 Morning Shift] Q#1 (MCQ):** "There are three co-centric conducting spherical shells A,B and C of radii a,b and c respectively ($c > b > a$) and they are charged wi..." **[Ans: D]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Two point charges $2q$ and q are placed at vertex A and centre of face CDEF of the cube as shown in figure. The electric flux passing ..." **[Ans: B]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Two point charges $q_1 = 3\mu\text{C}$ and $q_2 = -4\mu\text{C}$ are placed at points $(2\mathbf{i} + 3\mathbf{j} + 3\mathbf{k})$ and $(\mathbf{i} + \mathbf{j} + \mathbf{k})$ respectively. Force on charge q_2 is ____ N. (Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI Units)

- (A) $(12\mathbf{i} + 24\mathbf{j} + 24\mathbf{k}) \times 10^{-3}$
- (B) $(4\mathbf{i} + 8\mathbf{j} + 8\mathbf{k}) \times 10^{-3}$ **[Correct]**
- (C) $(3\mathbf{i} + 6\mathbf{j} + 6\mathbf{k}) \times 10^{-3}$
- (D) $(-4\mathbf{i} - 8\mathbf{j} - 8\mathbf{k}) \times 10^{-3}$

Correct Answer: B

Solution Analysis: Coulomb's Law states that the force between two point charges is proportional to the product of the charges and inversely proportional to the square of the distance between them. In vector form, the force $\text{vec}(\mathbf{F})_{(21)}$ exerted on charge q_2 by charge q_1 is: $\text{vec}(\mathbf{F})_{(21)} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\text{vec}(\mathbf{r})|^2} \text{vec}(\mathbf{r})$ Where, $\text{vec}(\mathbf{r}) = \text{vec}(\mathbf{r})_{(2)} - \text{vec}(\mathbf{r})_{(1)}$ is the position vector from q_1 to q_2 , $\mathbf{r} = \frac{\text{vec}(\mathbf{r})}{|\text{vec}(\mathbf{r})|}$ is the unit vector and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ Substituting the unit vector, we get: $\text{vec}(\mathbf{F})_{(21)} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\text{vec}(\mathbf{r})|^2} \text{vec}(\mathbf{r})$

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Morning Shift (Q#1)

There are three co-centric conducting spherical shells A, B and C of radii a, b and c respectively ($c > b > a$) and they are charged with charge q_1, q_2 and q_3 respectively. The potentials of the spheres A, B and C respectively, are :

- (A) $\frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{b} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{c} \right)$
- (B) $\frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{a} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{b} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{c} \right)$
- (C) $\frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{a} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right)$
- (D) $\frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{a} + \frac{q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2}{b} + \frac{q_3}{c} \right), \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 + q_2 + q_3}{c} \right)$ **[Correct]**

Correct Answer: D

Solution Analysis: The electric potential V due to a charged spherical shell of radius R and charge Q is given by :

$V = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{R} \right), r < R$ $\frac{1}{4\pi\epsilon_0} \left(\frac{Q}{R} \right), r = R$ $\frac{1}{4\pi\epsilon_0} \left(\frac{Q}{r} \right), r > R$ Point A is on the surface of shell A, inside shell B, and inside shell C. The total potential is the sum of potentials due to each shell (Superposition Principle). So, the potential at A ; 1. Due to A (Surface, $r=a$) $\Rightarrow V_{(AA)} = \frac{1}{4\pi\epsilon_0} \frac{q_1}{a}$ 2. Due to B (Inside, $r < b$) $\Rightarrow V_{(AB)} = \frac{1}{4\pi\epsilon_0} \frac{q_2}{b}$ 3. Due to C (I